Project 3 - Geostatistical Assessment of the Built Environment

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Background

The goal of this project is to perform a noise pollution assessment between an open space and a nearby neighborhood area. Balcones District Park was the chosen open space and the neighborhood of nearly the same area was chosen across Amherst Drive. Balcones District Park is in far northwest Austin at the northeast corner of Duval Road and Amherst Drive. It's a unique park that is heavily wooded in the eastern part and an open park with a playground, pool and basketball court in the western part. The neighborhood across the street is primarily single-family houses with many trees and medium sized yards. It is predicted that the areas near Amherst Drive and Duval Road will have the highest decibel readings and the areas farthest away from these streets will have the lowest readings. These readings will then be analyzed using both deterministic and geostatistical interpolation.

Methodology

Base Map Creation

The basemap used in this study was LiDAR and imagery collected from the TxGIO Data Hub. The imagery used is from the "CapArea & McLennan Imagery 2020" and the LiDAR is from "Central Texas 2017" section in TxGIO Data Hub. The tile index was also downloaded for both sets of files to find the correct files that overlap the study area. The LAS files from this were then converted to DEM files. With ArcGIS Pro, a raster mosaic was then created from the imagery files. Once the basemap was created. The routes for the data collection were then added to the basemap. The goal here was to gather enough datapoints around the interior and perimeter of both the residential and open space areas to be able adequately model the data via interpolation. The map was then uploaded to ArcGIS Online under the data gatherer's account within Austin Community College's site. Figure 1 shows the routes in the study area and Figure 2 shows the digital elevation model (DEM) of the study area.

Data Collection

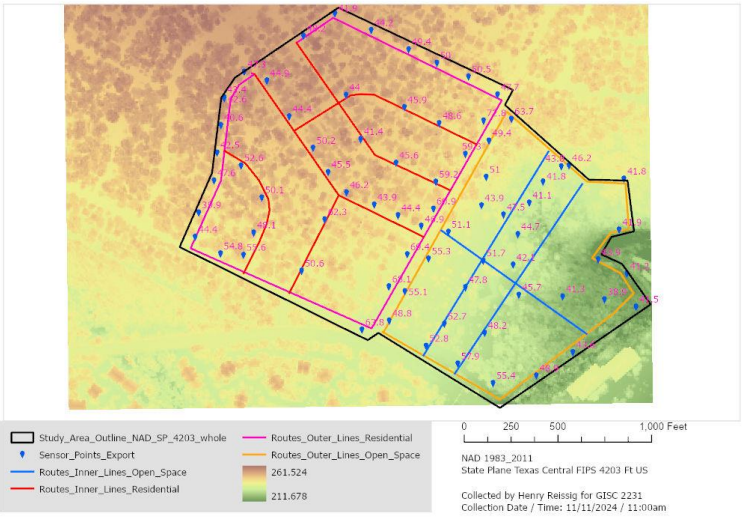
Data collection of the sound reading was done with the NIOSH SLM application on the iPhone and the recording of the data was done through ArcGIS Field Maps. LAq (A-weighted

Equivalent Continuous Sound Level) was used for 15 seconds for each reading. Care was taken to ensure that the sound detected is an ambient sound free from extraneous noises like airplanes, birds, leaf blowers, talking, etc. when possible. The data was collected on Monday November 11, 2024, from around 10:00am to 12:00pm. Figure 1 and Figure 2 show the study area along with the sound readings in both the neighborhood area and the open space area (Balcones District Park).

Figure1. Field Data Collection Routes



Figure 2. Field Data Collection Routes DEM



Analysis

The data points were added to the ArcGIS Pro project and a variety of deterministic and geospatial interpolation models were performed as discussed in the “Results” section of the project.

Figure 3 – Residential Area



Figure 4 – Open Space (Balcones District Park)



Results

Figure 5a shows graphically how near the distribution of the readings “Sensor_Reading_Adust” are to being a normal distribution line and Figure 5b shows the

distribution of the reading and where the mean is relative to the rest of the distribution. It's skewed towards the low end of the readings and stretches out towards the higher readings.

Figure 5(a)

Normal QQ Plot

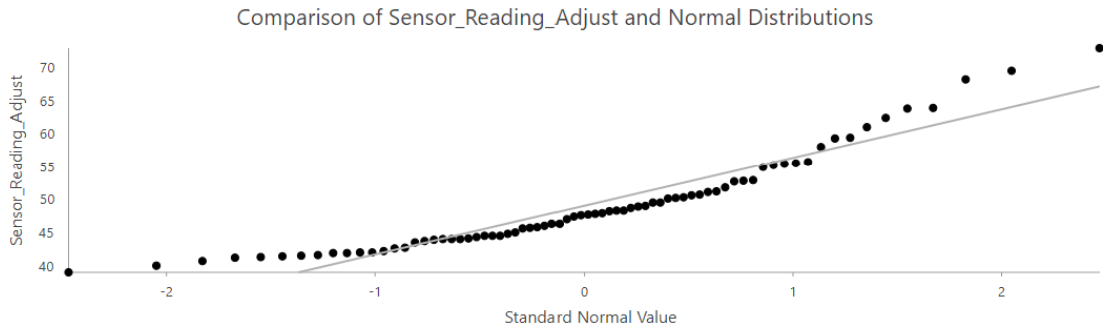
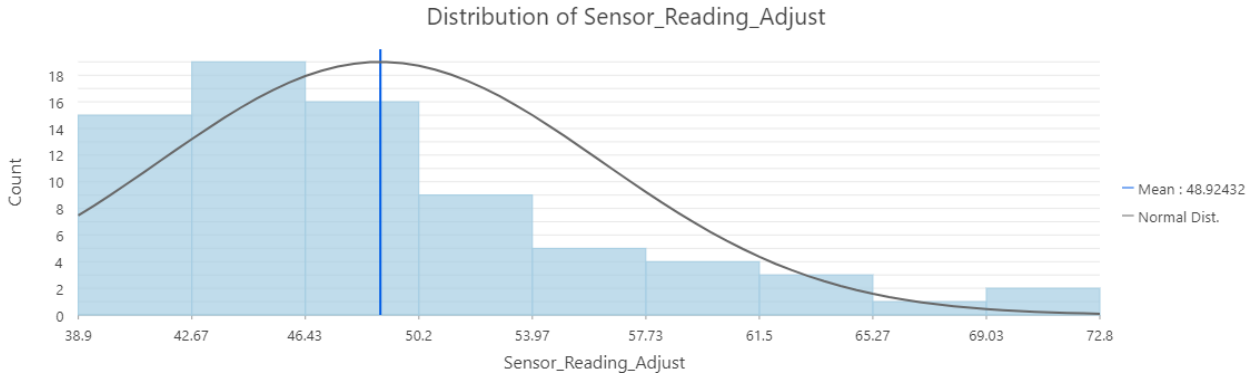


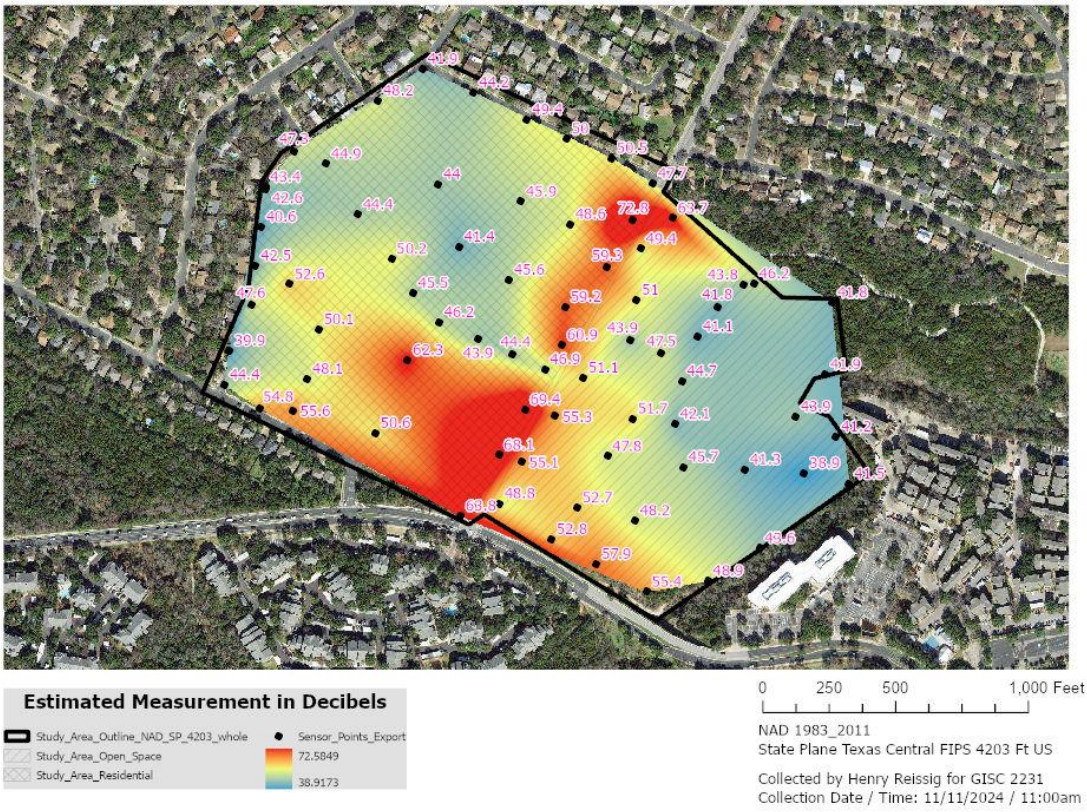
Figure 5(b)



Deterministic models

Three deterministic models were used to model the results: Inverse Distance Weighted (IDW), Spline Interpolation, and Natural Neighbor Interpolation. Figure 6 shows the results of natural neighbor interpolation. This model was selected because it best represented the initial expected results, and it best matched the tangible experience in the field. In other words, the areas in red near the streets felt noisier than those areas deep in the neighborhood or deep in the woods in the park.

Figure 6. Interpolated Decibel Readings



Geospatial interpolation models

Kriging was the geospatial interpolation model of choice for this project. Figure 7 shows how well the model predicts the current measurements. Empirical Bayesian Kriging was the chosen Kriging method. Of the three tried and examining the cross validation as in Figure 7 this model and the best overall values for the lowest RMS error, standardized RMS closest to 1 and a standard mean nearest to zero. Figure 8 shows the results of the Empirical Bayesian Kriging Model. Like the natural neighbor interpolation model in Figure 6, Figure 7 has very intuitive results because of correlation between street distance to high levels of noise and areas deep in the neighborhood and woods having the lowest values.

Figure 7

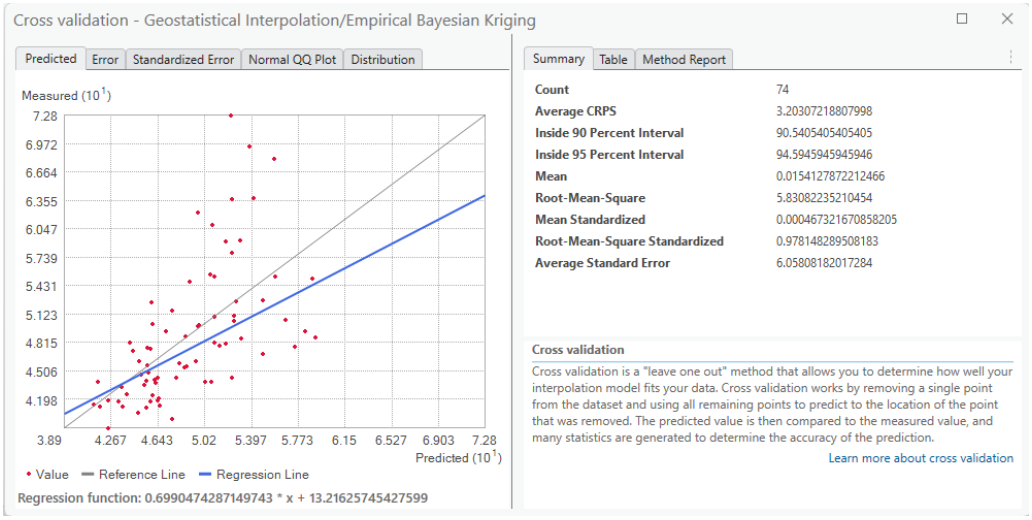


Figure 8. Kriging



Conclusion

The goal to perform a noise pollution assessment between an open space and a nearby neighborhood area for this project was met. The data routes were mapped, the maps were created, the sound measurements were made, and the analysis was performed in this project to thoroughly map out the interpolated layers that predicts potential sound measurement for the entire study area. As mentioned in the methodology section, both interpolation models (Natural Neighbor Interpolation and Empirical Bayesian Kriging Model) adequately represent reality in the field where the higher readings are along the streets and attenuate from there away from the streets. Anyone wanting to live in the neighborhood across the street will benefit from this analysis to find a home in the quietest area possible and familiarity with the sound interpolated layer for the park could benefit wildlife studies that could depend on sound levels.